# CPE 381 Classwork 09: Time-invariance, Period of a Signal

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 23, 2026 (In-Class)

1. Is the system described by input-output relationship $y(t) = x(t) \cdot \cos(2\pi t)$ a time-invariant system? Show your work. **(10 Points)**

**Answer**

2. Find the fundamental period of $x(t) = \cos(2\pi t) + \sin(4\pi t) + \cos(6\pi t)$. **(10 Points)**

**Answer**